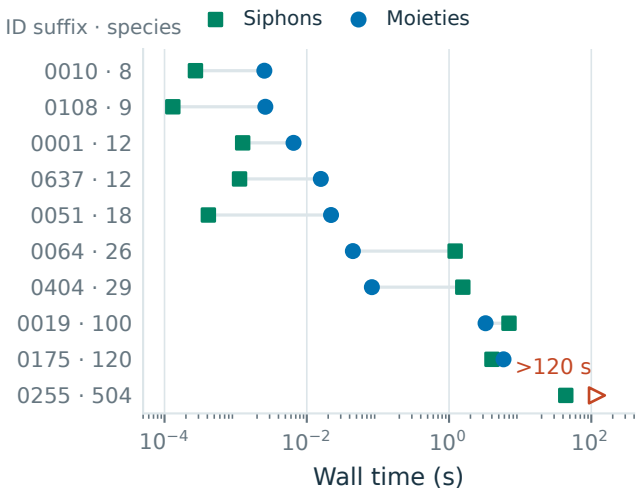
A Reference checks and independent SBML comparison

	Species	Rxns	n	ℓ	s	δ	Laws	WR	Tests	External
Reversible isomerization	2	2	2	1	1	0	1	●	✓	8/8
Irreversible isomerization	2	1	2	1	1	0	1	○	✓	8/8
Isomerization cycle	3	3	3	1	2	0	1	●	✓	8/8
Disconnected reversible pairs	6	6	6	3	3	0	3	●	✓	8/8
Pairs with stoichiometry	6	6	6	3	3	0	3	●	✓	8/8
Michaelis–Menten	4	3	3	1	2	0	2	○	✓	8/8
Reversible Michaelis–Menten	4	4	3	1	2	0	2	●	✓	8/8
Edelstein switch	3	6	5	2	2	1	1	●	✓	8/8
Phosphorylation: one site	6	6	6	2	3	1	3	○	✓	8/8
Phosphorylation: two sites	9	12	10	2	6	2	3	○	✓	8/8
Shinar–Feinberg motif	2	2	4	2	1	1	1	○	✓	8/8
Open reactor	2	3	3	1	2	0	0	●	✓	8/8

Colour scaled within each count column; WR: ● yes / ○ no; external: shared quantities agreeing.

B BioModels analysis costs



10/10 supported structural round trips

C Conserved pools in KEGG

One circle = one minimal conserved pool

Glycolysis M00001
14 species · 15 reactions · δ = 1
11 2 1

Pyruvate oxidation M00307
13 species · 8 reactions · δ = 1
5 4 3 2 2 2 2 2

Citrate cycle M00009
22 species · 12 reactions · δ = 0
8 8 3 3 3 2 2 2 2 2 2 2

Pentose phosphate M00004
12 species · 12 reactions · δ = 0
11 11

Numbers = species in each pool